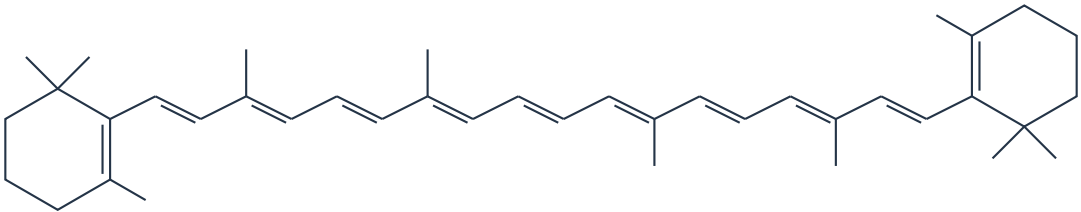
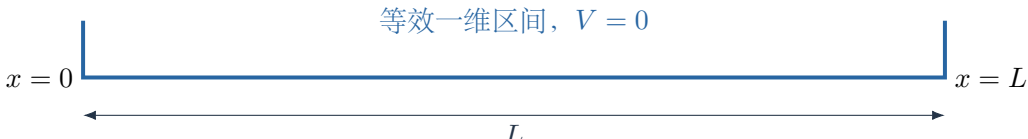
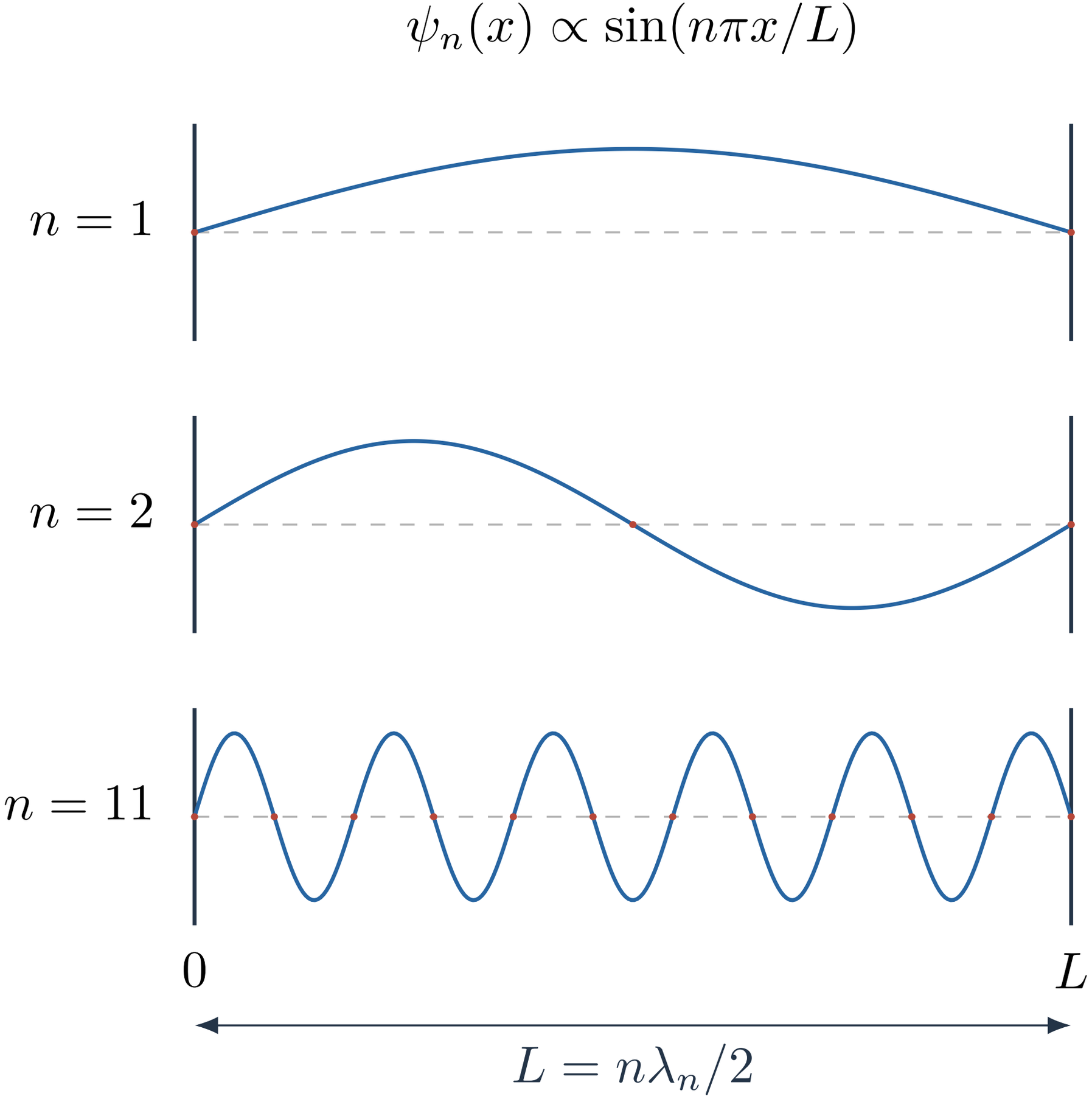
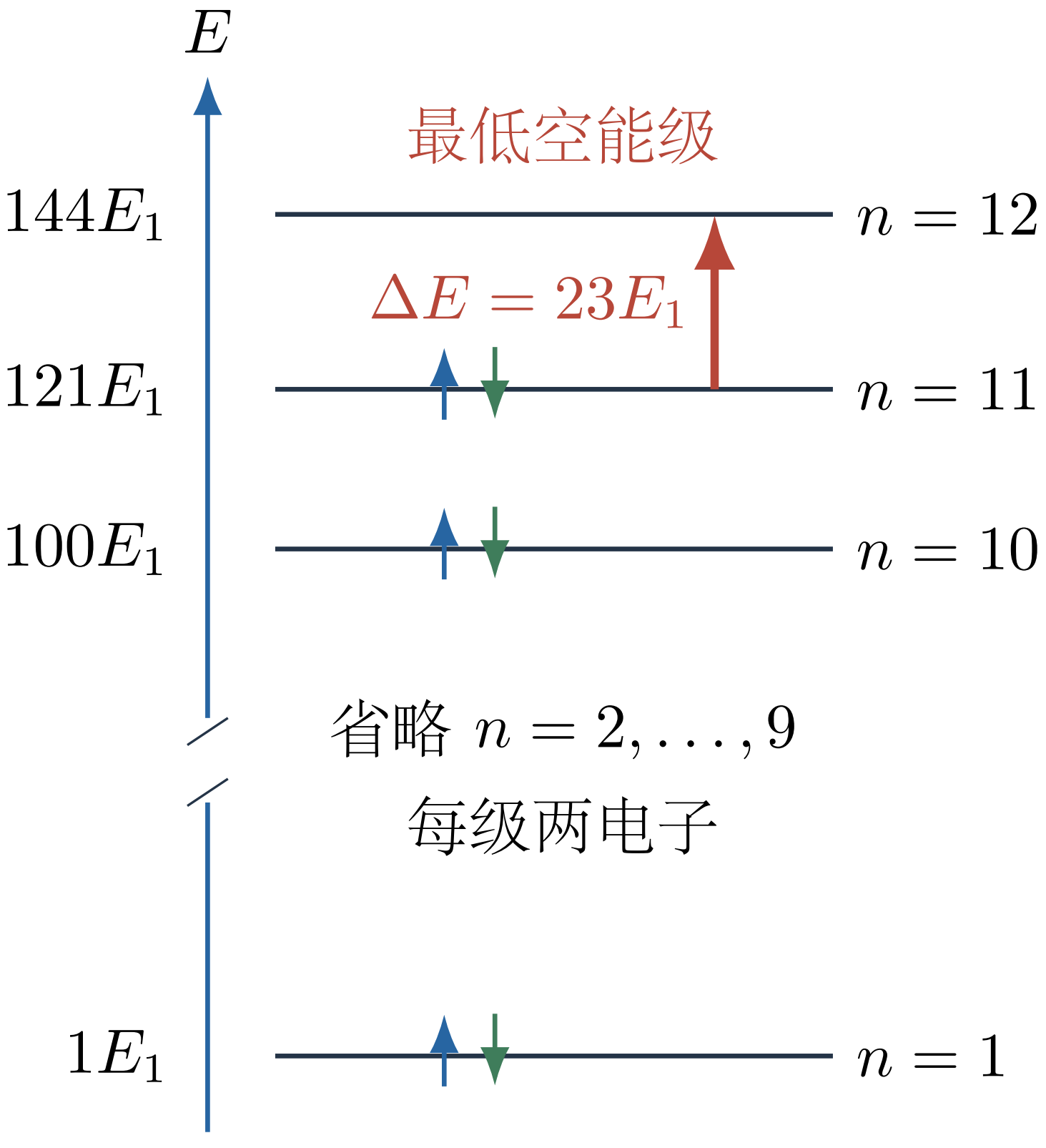
β-胡萝卜素：$C_{40}H_{56}$






$\psi_n(x) \propto \sin(n\pi x/L)$
$n = 1$
$n = 2$
$n = 11$
0
L
$L = n\lambda_n/2$


E
最低空能级
$144E_1$
$n = 12$
$\Delta E = 23E_1$
$121E_1$
$n = 11$
$100E_1$
$n = 10$
省略 $n = 2, \ldots, 9$
每级两电子
$1E_1$
$n = 1$
22 电子填满 $n = 1$ 至 11